

Aman Katyal

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Education

Purdue University

- Bachelor of Science in Computer Engineering | **GPA: 3.91** , Dean's List / Semester Honors Expected May 2027
- Master of Science in Electrical and Computer Engineering Expected May 2028
- **Relevant Coursework:** RISC-V Microarchitecture, ASIC Design Flow, Embedded Microprocessor Systems

Experience

Hewlett Packard Enterprise | ASIC Intern | Roseville, CA

May 2026 – Aug 2026

- Rebuilt an internal microcode compilation pipeline in Python with Lark, defining an EBNF grammar, AST validator, and CLI toolchain that compiles network tracking rules into on-chip instructions
- Integrated a binary disassembler and instruction visualizer into the toolchain to validate generated microcode against hardware execution constraints
- Built a pre-silicon log ingestion engine in DuckDB that parses 10 GB of simulation traces in under 2 minutes into relational schemas indexed by cycle timestamp and transaction ID, isolating pipeline stalls and buffer bottlenecks
- Deployed a Streamlit dashboard that tracks RTL Git commit SHAs and runs differential SQL queries on port bandwidth and memory utilization, speeding up post-failure triage

Purdue System on Chip Extension Technologies (SoCET) | Undergraduate ASIC Researcher

Aug 2025 – Present

- Profile compiler-generated kernel assembly against hand-written benchmarks in a functional simulator for an AI accelerator, identifying redundant instructions, register spills, and runtime gaps to guide compiler optimizations
- Develop VLIW packetization and software pipelining passes in the PPCI compiler for the AI accelerator's multi-slot ISA, overlapping loop iterations to raise packet issue-slot utilization
- Ported TinyUSB to a custom RISC-V SoC, implementing a bare-metal USB Host Controller Driver with a PLIC-driven ISR and validating data flow through Verilator co-simulation

Projects

Purdue IEEE ROV Team: Multi-Node Vehicle Network | Embedded Lead

May 2026 – Present

- Migrated vehicle communication from point-to-point SPI to a multi-node CAN FD bus (1 Mbps nominal, 5 Mbps data), enabling direct messaging across 3 STM32 microcontrollers
- Built a software-in-the-loop simulation dashboard to validate multi-node firmware telemetry and pilot controls on host machines before vehicle assembly
- Designed a 4-layer KiCad PCB for HIL validation and RP2350B vs. STM32 benchmarking, decoupling firmware development from vehicle manufacturing and reducing the design cycle by 12 weeks

Autonomous Profiling Module

Jan 2026 – May 2026

- Wrote bare-metal C firmware on an RP2040 with I2C sensor drivers (256x oversampling, bus recovery) driving a 10 Hz depth PID loop that holds depth within 5 cm
- Designed a compact binary telemetry protocol over SPI LoRa transceivers, enabling wireless PID tuning without opening the sealed hull and persisting tuned gains to flash

Harry Potter Wand Duel: Laser-Tag-Style Game with ML Gestures

Aug 2025 – Dec 2025

- Programmed an RP2350B wand to classify spell gestures with a quantized 5.3 KB TensorFlow Lite Micro model (96% accuracy, under 5 ms latency), isolating 1 kHz IMU sampling and inference from game logic across both cores via SIO FIFOs
- Engineered a wearable chestpiece with a custom NEC infrared transceiver, PIO-driven RGB matrix, and haptic feedback for real-time hit validation

Skills

Languages: C, C++, Python, SystemVerilog, SQL, Assembly, Bash

Embedded Systems: Device Drivers, HAL, FreeRTOS, TinyUSB, CAN FD, SPI, I2C, UART, PWM, DMA, ADC, PIO, TinyML (TensorFlow Lite Micro)

Hardware and Lab: KiCad, LTspice, STM32, RP2350, RP2040, ARM Cortex-M, RISC-V, Logic Analyzers, Oscilloscopes

Tools: Git, Verilator, UVM, Lark, DuckDB, Streamlit